Introduction to Data Analytics
MS - 4160

Session 9 – Regression Revised

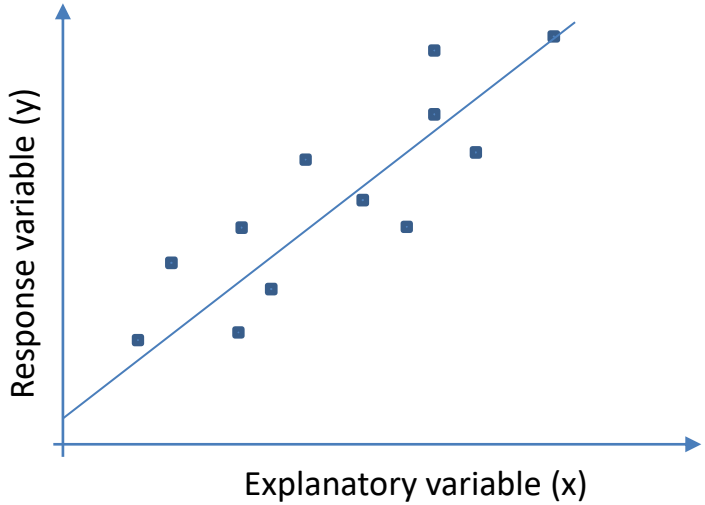
Nandan Sudarsanam,
Department of Management Studies,
Robert Bosch Centre for Data Science and AI (RBC-DSAI),
Indian Institute of Technology Madras

A review of regression

- Regression Analysis is the study of relationships between variables
  - Going beyond categorical variables
  - Model based relationship (linear and non-linear)
  - Useful towards interpretation and prediction
- Examples
  - Relationship between speed and fuel efficiency of a Truck; Operating drill speed and soil/stone density affect on rate of drill
  - How do wages of employees get affected with experience, education, promotions, etc.
  - How does sales revenue get affected as a function advertising expenses, competitors advertisements, etc.
  - How does the price of a property get affected by square footage, distance from city centre, etc.
  - How does the current price of a currency exchange rate depend on its past values

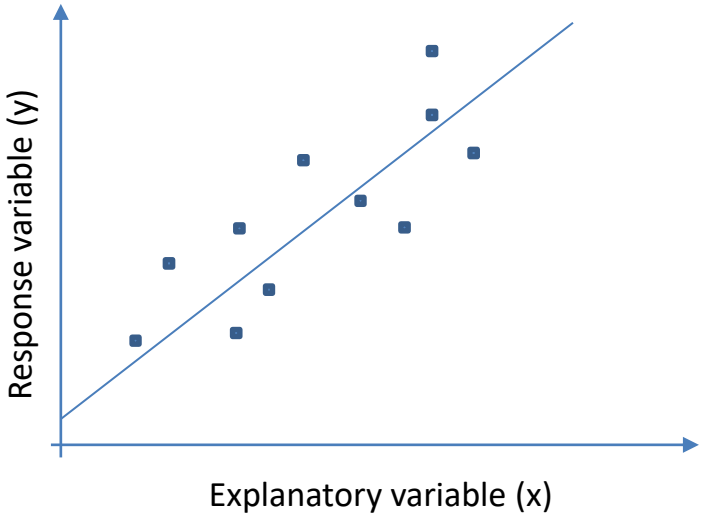
Categorizations, nomenclature and concepts

- Linear versus non-linear
- Simple versus multiple
- Cross-sectional versus time-series
- Response (or Dependent) variables
- Explanatory (or Independent) variables
- Scatter plots and outliers
- Correlations (a quantitative indicator of linear relationship)
  - R^2 is correlation square
  - Covariance normalized
  - Can covariance be theoretical and empirical? Also, what is the covariance matrix



Fitting lines

- Role of descriptive and inferential statistics
- y= mx+c or y = b_0+b_1x
- Optimization to identify the best fit
- Inference on the individual parameters
  - The t-test for the H_0: $\beta_1 = 0$
- Inference for the overall model
  - ANOVA of a different kind



Fitting lines

- Linear model between y_i and x_i , $i = 1,\ldots, n$

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

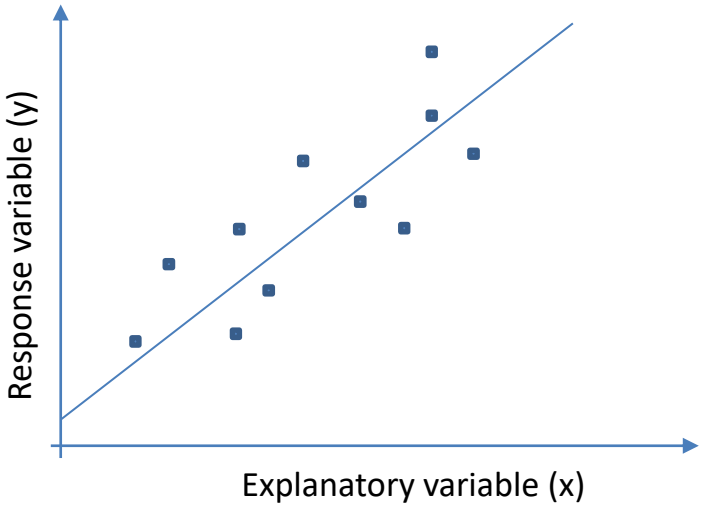
- Therefore error

$$\epsilon_i = y_i - \beta_0 - \beta_1 x_i$$

- The sum of squares of errors (SSE)

$$\sum_i \epsilon_i^2 = \sum_i (y_i - \beta_0 - \beta_1 x_i)^2$$

- The minimization of SSE gives estimates of β_0 and β_1 in closed form
- The $R^2 = \frac{SST - SSE}{SST}$



Statistical Inference

- We get some $\hat{\beta}_1$, but we are interested in β_1
- So
  - CI $\left[\hat{\beta}_1 - 2 \cdot \mathrm{SE}(\hat{\beta}_1),\ \hat{\beta}_1 + 2 \cdot \mathrm{SE}(\hat{\beta}_1)\right]$
  - Hypothesis tests: $H_0 : \beta_1 = 0$ and $t = \dfrac{\hat{\beta}_1 - 0}{\mathrm{SE}(\hat{\beta}_1)}$
  - Where

$$\mathrm{SE}(\hat{\beta}_1)^2 = \frac{\sigma^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

Assumes independence

Some important questions when extending to multiple regression

- From ISLR

1. *Is at least one of the predictors* $X_1, X_2, \ldots, X_p$ *useful in predicting the response?*

2. *Do all the predictors help to explain* Y*, or is only a subset of the predictors useful?*

3. *How well does the model fit the data?*

4. *Given a set of predictor values, what response value should we predict, and how accurate is our prediction?*

Some Answers

1) Just like we did $H_0 : \beta_1 = 0$ why not get p-values for each beta?
   a) Type 1 error
   b) Multicollinearity

Instead we $H_0 : \beta_1 = \beta_2 = \cdots = \beta_p = 0$ vs $H_a :$ at least one β_j is non-zero.

By $F = \frac{(SST - SSE)/p}{SSE/(n-p-1)}$

2) & 3) Why not drop anything with a high p-value?
We need a global measure of quality of fit.

Mallow's C_P, AIC, BIC, Adjusted R^2, p-value of F-stat, pretty soon we are going to talk about CV error

Some answers

2)&3) continued: Ok you have a global measure now what?

a) Forward selection, b) Backward selection, c) Mixed/Hybrid, d) Best subsets

4) Remember we had the $\mathrm{SE}(\hat{\beta}_1)$ We translated this to CI. We can propagate these to the response. In which case do you have the range of possible output values? No. This is only the reducible component. It translates to a CI on the response which is an interval for the average response. You need to include the irreducible component to get a prediction interval of the response.

Potential problems

- Potential problems (ISLR)

1. *Non-linearity of the response-predictor relationships.*
2. *Correlation of error terms.*
3. *Non-constant variance of error terms.*
4. *Outliers.*
5. *High-leverage points.*
6. *Collinearity.*

Some Answers

1) Non-linear transformations and interaction terms. Scatter plots and plots of residuals help
2) & 3) Plots of residuals and standardized residuals help. Transformations of the input space help.

6) What is multicollinearity

VIF

– Variance Inflation Factor (VIF) $Var(\hat{\beta}_j) = \frac{1}{1-R_j^2} \frac{\hat{\sigma}^2 c_{jj}}{(n-1)}$

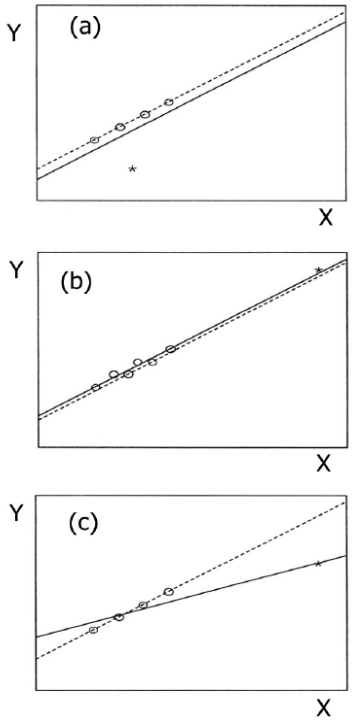
– consequences on interpretation vs prediction vs stability

– Dimensionality reduction and regularization

Some answers

- Answers for 4&5)
- Lecture borrows from: http://polisci.msu.edu/jacoby/icpsr/regress3/lectures/week3/11.Outliers.pdf

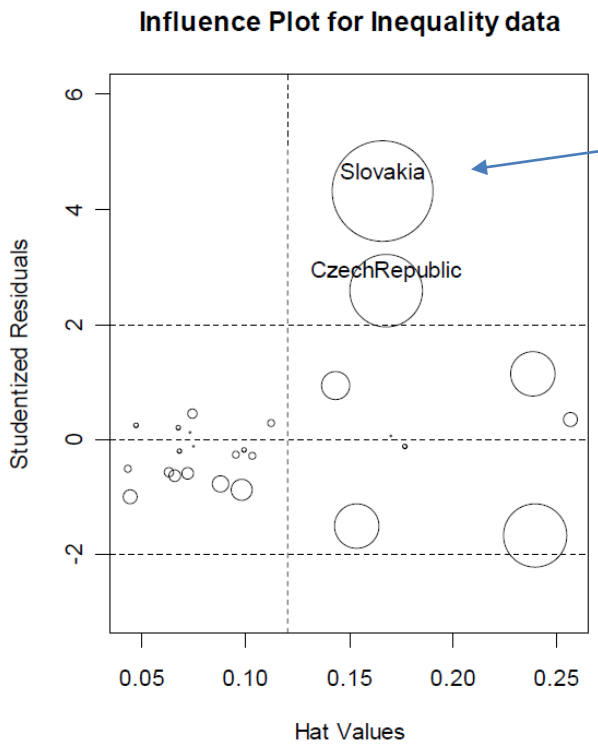
  - Outlier without influence: Studentized residual catches it
  - High leverage: (also called hat values) will be high. But we should not flag this
  - High leverage: Pii value is high but Studentized residual is not, and we should flag this
  - What about DFBetas/CooksD?



Adapted from Figure 11.1 (Fox, 1997)

Some answers

- We need an Influence or Bubble plot



Size of circle based on Cooks D